Subspaces of $\mathbb{R}^n$

A set H of vectors in $\mathbb{R}^n$ is called a subspace if :

1. The zero vector $\mathbf{0}$ is in H.
2. For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is also in H.
3. For each vector $\mathbf{u}$ in H and every scalar c, the vector $c\mathbf{u}$ is also in H.

Basis of a subspace H of $\mathbb{R}^n$

Suppose H is a subspace of $\mathbb{R}^n$. A list of vectors $\mathbf{a}_1, ..., \mathbf{a}_k$ in $\mathbb{R}^n$ is called a basis of H if:

1. $\mathbf{a}_1, ..., \mathbf{a}_k$ are in H.
2. $\text{Span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} = H$.
3. $\mathbf{a}_1, ..., \mathbf{a}_k$ are linearly independent.

Basis for Col A

Col A

Suppose A is an $m \times n$ matrix. We can write A as

$$A = [\ \mathbf{a}_1\ \mathbf{a}_2\ \cdots\ \mathbf{a}_n\ ]$$

Col A is the subspace of $\mathbb{R}^m$ spanned by the columns of A.

$$\text{Col}\, A = \text{Span}\{\mathbf{a}_1, \cdots, \mathbf{a}_n\}.$$

Algorithm for finding a basis of Col A

Suppose A is an $m \times n$ matrix. To find Col A:

1. Find the R.E.F. (or just the E.F.) of A.
2. The pivot columns of A will be a basis of Col A.

Example

Example

Find a basis for $\operatorname{Col} A$ where $A = \begin{bmatrix} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{bmatrix}$

Solution. First we row reduce A :

$$\begin{bmatrix} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{bmatrix} \xrightarrow{\text{Row Reduction}} \begin{bmatrix} 1 & -2 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ R.E.F.}$$

The pivot columns of A are $\mathbf{a}_1$ and $\mathbf{a}_3$. Therefore a basis for $\operatorname{Col} A$ is given by:

$$\{\mathbf{a}_1, \mathbf{a}_3\} = \left\{ \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} \right\}$$

Basis for Nul A

Nul A

Suppose A is an $m \times n$ matrix.

The null space of A is the subspace of $\mathbb{R}^n$ containing all $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{0}$.

The null space of A is denoted by Nul A.

Algorithm for finding a basis of Nul A

1. Solve the linear system $A\mathbf{x} = \mathbf{0}$ in the usual way, i.e., by finding R.E.F. of $[\ A \mid 0\ ]$.
2. Write the solution in vector parametric form :

$$\mathbf{x} = x_{i_1}\mathbf{c}_1 + \cdots + x_{i_r}\mathbf{c}_r$$

3. The vectors $\mathbf{c}_1, \ldots, \mathbf{c}_r$ will be a basis of Nul A.

Example

Find a basis for $\operatorname{Nul} A$ where $A = \begin{bmatrix} 1 & -1 & 0 & -2 & 0 \\ 2 & -2 & 1 & -1 & 0 \\ 1 & -1 & -1 & -5 & 1 \end{bmatrix}$

Solution. To solve the system $A\mathbf{x} = \mathbf{0}$ we need to find the R.E.F. of the following augmented matrix :

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & -2 & 0 & 0 \\ 2 & -2 & 1 & -1 & 0 & 0 \\ 1 & -1 & -1 & -5 & 1 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccccc|c} 1 & -1 & 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

x_1, x_3, x_5 : basic
x_2, x_4 : free

$$\begin{cases} x_1 = x_2 + 2x_4 \\ x_3 = -3x_4 \\ x_5 = 0 \end{cases} \implies \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} x_2 + 2x_4 \\ x_2 \\ -3x_4 \\ x_4 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ 0 \\ -3 \\ 1 \\ 0 \end{bmatrix}$$

And a basis for $\operatorname{Nul} A$ is given by $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -3 \\ 1 \\ 0 \end{bmatrix} \right\}$

Example

Find a basis for $\operatorname{Nul} A$ where $A = \begin{bmatrix} 1 & -2 & -1 & -1 \\ -2 & 4 & 1 & -1 \\ 2 & -4 & 0 & 4 \\ 1 & -2 & -2 & -4 \end{bmatrix}$

Solution. To solve the system $A\mathbf{x} = \mathbf{0}$ we need to find the R.E.F. of the following augmented matrix :

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & -1 & 0 \\ -2 & 4 & 1 & -1 & 0 \\ 2 & -4 & 0 & 4 & 0 \\ 1 & -2 & -2 & -4 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{cccc|c} 1 & -2 & 0 & 2 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

x_1, x_3 : basic
x_2, x_4 : free

$$\begin{cases} x_1 = 2x_2 - 2x_4 \\ x_3 = -3x_4 \end{cases} \implies \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2x_2 - 2x_4 \\ x_2 \\ -3x_4 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -2 \\ 0 \\ -3 \\ 1 \end{bmatrix}$$

And a basis for $\operatorname{Nul} A$ is given by $\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ -3 \\ 1 \end{bmatrix} \right\}$

Theorem

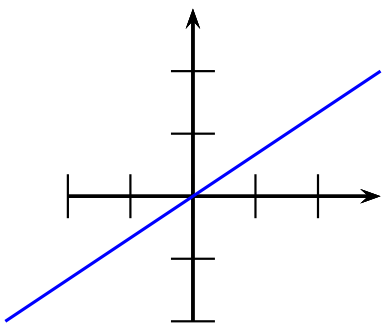
Every two bases of a vector space H have the same number of vectors.

Dimension of a subspace H

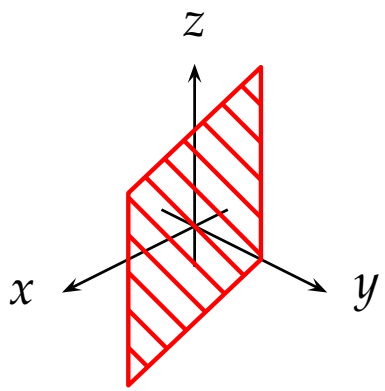
Suppose H is a subspace of $\mathbb{R}^n$.

The number of vectors in any basis of H is called the dimension of H.

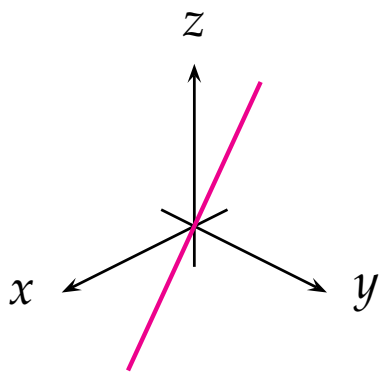
Remark. By convention, the subspace $\{\mathbf{0}\}$ has dimension zero.



$\dim H = 1$



$\dim H = 2$



$\dim H = 1$

Rank of a matrix

Suppose A is an $m \times n$ matrix. The rank of A is the dimension of $\operatorname{Col} A$.

We denote the rank of A by "rank A".

- $\operatorname{rank} A = \dim \operatorname{Col} A =$ number of pivot columns of A.
- $\dim \operatorname{Nul} A =$ number of free variables of $A\mathbf{x} = \mathbf{0}$
  $=$ number of non-pivot columns of A.

The Rank Theorem

Suppose A is an $m \times n$ matrix. Then :

$$\operatorname{rank} A + \dim \operatorname{Nul} A = n$$

Sometimes $\dim \operatorname{Nul} A$ is called the nullity of A.

Example

Let $A = \begin{bmatrix} 1 & -1 & 0 & -2 & 0 \\ 2 & -2 & 1 & -1 & 0 \\ 1 & -1 & -1 & -5 & 1 \end{bmatrix}$.

Earlier today we saw that :

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & -2 & 0 & 0 \\ 2 & -2 & 1 & -1 & 0 & 0 \\ 1 & -1 & -1 & -5 & 1 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccccc|c} 1 & -1 & 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

Therefore $\operatorname{rank} A = 3$.

We also saw that a basis for $\operatorname{Nul} A$ is given by $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -3 \\ 1 \\ 0 \end{bmatrix} \right\}$

Therefore $\dim \operatorname{Nul} A = 2$.

Observe that $\operatorname{rank} A + \dim \operatorname{Nul} A = 3 + 2 = 5$= number of columns of A.

Inverse of a matrix

Suppose A is an $n \times n$ matrix. We say that A is invertible if there exists an $n \times n$ matrix C such that

$$AC = CA = \mathrm{I}_n$$

The matrix C is called the inverse of A, and is usually denoted by A^{-1}.

Theorem (2×2 matrices)

Suppose $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then :

- If $ad - bc = 0$ then A is not invertible.
- If $ad - bc \neq 0$ then A is invertible and we have

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- The quantity $ad - bc$ is called the determinant of A.

Determinant of an $n \times n$ matrix

- Our goal is to obtain a determinant for an $n \times n$ matrix, whose main property is that :

$$\boxed{\det A \neq 0 \text{ if and only if } A \text{ is invertible}}$$

Determinant of a 2×2 matrix

If $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then

$$\det A = a_{11}a_{22} - a_{12}a_{21}$$

We write this as $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$

Cofactors

Let A be an $n \times n$ matrix. Suppose $1 \leqslant i \leqslant n$ and $1 \leqslant j \leqslant n$ are integers.

The (i,j)-minor of A is the $(n-1) \times (n-1)$ matrix obtained by eliminating the i-th row and the j-th column of A. It is denoted by A_{ij}.

Example. $A = \begin{bmatrix} 1 & -4 & 5 \\ 0 & 2 & 1 \\ 3 & 2 & 4 \end{bmatrix}$ then :

$$A_{31} = \begin{bmatrix} -4 & 5 \\ 2 & 1 \end{bmatrix}$$

$$A_{11} = \begin{bmatrix} 2 & 1 \\ 2 & 4 \end{bmatrix}$$

$$A_{22} = \begin{bmatrix} 1 & 5 \\ 3 & 4 \end{bmatrix}$$

$$A_{23} = \begin{bmatrix} 1 & -4 \\ 3 & 2 \end{bmatrix}$$

Determinant of a 3×3 matrix

Suppose $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$.

The determinant of A is the following expression :

$$\det A = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

- This formula is called the **cofactor expansion across the first row** of A.
- We can also write it as

$$\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + a_{13} \det A_{13}$$

Example Suppose $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 3 & 2 \\ -2 & 1 & 0 \end{bmatrix}$. Then

$$\det A = 1 \begin{vmatrix} 3 & 2 \\ 1 & 0 \end{vmatrix} - (-1) \begin{vmatrix} 0 & 2 \\ -2 & 0 \end{vmatrix} + 0 \begin{vmatrix} 0 & 3 \\ -2 & 1 \end{vmatrix} = 1(-2) - (-1)(4) + 0(6) = 2$$

Determinant of a 3×3 matrix - other row expansions

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

- **Cofactor expansion across the first row :**

$$\det A = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

- **Cofactor expansion across the second row :**

$$\det A = -a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{22} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{23} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$$

- **Cofactor expansion across the third row :**

$$\det A = a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} - a_{32} \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} + a_{33} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

Question. How are the $\pm$ signs chosen?

Answer According to $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

Example

Suppose $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 3 & 2 \\ -2 & 1 & 0 \end{bmatrix}$. Then : $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

- Cofactor expansion across the first row :

$$\det A = 1\begin{vmatrix} 3 & 2 \\ 1 & 0 \end{vmatrix} - (-1)\begin{vmatrix} 0 & 2 \\ -2 & 0 \end{vmatrix} + 0\begin{vmatrix} 0 & 3 \\ -2 & 1 \end{vmatrix} = -2 + 4 + 0 = 2$$

- Cofactor expansion across the second row :

$$\det A = -0\begin{vmatrix} -1 & 0 \\ 1 & 0 \end{vmatrix} + 3\begin{vmatrix} 1 & 0 \\ -2 & 0 \end{vmatrix} - 2\begin{vmatrix} 1 & -1 \\ -2 & 1 \end{vmatrix} = -0 + 0 - 2(-1) = 2$$

- Cofactor expansion across the third row :

$$\det A = -2\begin{vmatrix} -1 & 0 \\ 3 & 2 \end{vmatrix} - 1\begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} + 0\begin{vmatrix} 1 & -1 \\ 0 & 3 \end{vmatrix} = (-2)(-2) - 2 + 0 = 2$$

One can also expand the determinant down any column :

Determinant of a 3×3 matrix - column expansions

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

- Cofactor expansion down the first column :

$$\det A = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

- Cofactor expansion across the second column :

$$\det A = -a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{22} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{32} \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$$

- Cofactor expansion across the third column :

$$\det A = a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} - a_{23} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} + a_{33} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & 22 \end{vmatrix}$$

Example

Suppose $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 3 & 2 \\ -2 & 1 & 0 \end{bmatrix}$. Then : $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

- Cofactor expansion down the first column :

$$\det A = 1\begin{vmatrix} 3 & 2 \\ 1 & 0 \end{vmatrix} - 0\begin{vmatrix} -1 & 0 \\ 1 & 0 \end{vmatrix} + (-2)\begin{vmatrix} -1 & 0 \\ 3 & 2 \end{vmatrix} = -2 - 0 + 4 = 2$$

Determinant of an $n \times n$ matrix

Suppose $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$ is an $n \times n$ matrix.

- The determinant of A can be computed by cofactor expansion across a row or down a column.
- The formula for the determinant of an $n \times n$ matrix depends on the formula for the determinant of an $(n-1) \times (n-1)$ matrix.

(i,j)-Cofactors

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

- Suppose A_{ij} denotes the (i,j)-minor of A. The (i,j)-cofactor of A is:

$$C_{ij} = (-1)^{i+j} \det A_{ij}$$

In other words:

$$C_{ij} = \begin{cases} \det A_{ij} & \text{if } i+j \text{ is even,} \\ -\det A_{ij} & \text{if } i+j \text{ is odd.} \end{cases}$$

The sign $(-1)^{i+j}$ is determined by the following checkerboard pattern of signs :

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \end{bmatrix}$$

E.g., $C_{11} = \det A_{11}$, $C_{12} = -\det A_{12}$, $C_{32} = -\det A_{32}$, etc.

Determinant of an $n \times n$ matrix

Suppose $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$ is an $n \times n$ matrix.

- **Cofactor expansion across the p-th row :**

$$\det A = a_{p1}C_{p1} + a_{p2}C_{p2} + \cdots + a_{pn}C_{pn}$$

- **Cofactor expansion across the q-th column :**

$$\det A = a_{1q}C_{1q} + a_{2q}C_{2q} + \cdots + a_{nq}C_{nq}$$

Example : 4×4 matrices

Let us consider 4×4 matrices:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

Cofactor expansion with respect to the 1st row:

$$\begin{aligned} \det A &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} + a_{14}C_{14} \\ &= a_{11} \det A_{11} - a_{12} \det A_{12} + a_{13} \det A_{13} - a_{14} \det A_{14} \end{aligned}$$

Cofactor expansion with respect to the 2nd row:

$$\begin{aligned} \det A &= a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23} + a_{24}C_{24} \\ &= -a_{21} \det A_{21} + a_{22} \det A_{22} - a_{23} \det A_{23} + a_{24} \det A_{24} \end{aligned}$$

Example

Compute the determinant of $A = \begin{bmatrix} 3 & 0 & 1 & 2 \\ -2 & 1 & 0 & 2 \\ 1 & 3 & 4 & 2 \\ -3 & 2 & 1 & -1 \end{bmatrix}$ using expansion down the third column.

Solution. The formula is

$$\det A = a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33} + a_{43}C_{43} = 1 \cdot C_{13} + 0 \cdot C_{23} + 4 \cdot C_{33} + 1 \cdot C_{43}$$

$$C_{ij} = (-1)^{i+j} \det A_{ij}$$

Using the following checkerboard pattern of signs :

$$\begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$$

we can write the formula as :

$$\det A = 1 \begin{vmatrix} -2 & 1 & 2 \\ 1 & 3 & 2 \\ -3 & 2 & -1 \end{vmatrix} - 0 \begin{vmatrix} 3 & 0 & 2 \\ 1 & 3 & 2 \\ -3 & 2 & -1 \end{vmatrix} + 4 \begin{vmatrix} 3 & 0 & 2 \\ -2 & 1 & 2 \\ -3 & 2 & -1 \end{vmatrix} - 1 \begin{vmatrix} 3 & 0 & 2 \\ -2 & 1 & 2 \\ 1 & 3 & 2 \end{vmatrix}$$

Example (cont.)

$$\begin{aligned}\det A &= 1\begin{vmatrix}-2 & 1 & 2\\ 1 & 3 & 2\\ -3 & 2 & -1\end{vmatrix} - 0\begin{vmatrix}3 & 0 & 2\\ 1 & 3 & 2\\ -3 & 2 & -1\end{vmatrix} + 4\begin{vmatrix}3 & 0 & 2\\ -2 & 1 & 2\\ -3 & 2 & -1\end{vmatrix} - 1\begin{vmatrix}3 & 0 & 2\\ -2 & 1 & 2\\ 1 & 3 & 2\end{vmatrix}\\ &= 31 - 0 + 4(-17) + (-1)(-26) = -11\end{aligned}$$

Because we can calculate each of the 3×3 determinants. For example :

$$\begin{vmatrix}-2 & 1 & 2\\ 1 & 3 & 2\\ -3 & 2 & -1\end{vmatrix} = (-2)\begin{vmatrix}3 & 2\\ 2 & -1\end{vmatrix} - 1\begin{vmatrix}1 & 2\\ 2 & -1\end{vmatrix} + (-3)\begin{vmatrix}1 & 2\\ 3 & 2\end{vmatrix} = (-2)(-7) - (-5) + (-3)(-4) = 31$$

(The others are left as an exercise.)

Example

Compute the determinant of $A = \begin{bmatrix} 1 & 2 & 5 & 1 \\ 0 & 0 & -1 & 0 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 0 \end{bmatrix}$

- It is generally easier to write an expansion for a row or column with lots of zeros.

$$\begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$$

$$\det A = 0 \cdot C_{21} + 0 \cdot C_{22} + (-1) \cdot C_{23} + 0 \cdot C_{24} = (-1)\Big(-\det(A_{23})\Big) = -(-1)\begin{vmatrix} 1 & 2 & 1 \\ 3 & 1 & 2 \\ 2 & 4 & 0 \end{vmatrix} = 10$$

because

$$\begin{vmatrix} 1 & 2 & 1 \\ 3 & 1 & 2 \\ 2 & 4 & 0 \end{vmatrix} = 1\begin{vmatrix} 1 & 2 \\ 4 & 0 \end{vmatrix} - 2\begin{vmatrix} 3 & 2 \\ 2 & 0 \end{vmatrix} + 1\begin{vmatrix} 3 & 1 \\ 2 & 4 \end{vmatrix} = 1(-8) - 2(-4) + 10 = 10$$

Triangular matrices

- An $n \times n$ matrix is called upper triangular if every entry below the diagonal is zero.

$$\begin{bmatrix} -1 & 3 & -1 & 3 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & -2 & 5 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

- An $n \times n$ matrix is called lower triangular if every entry above the diagonal is zero.

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 2 & 0 \\ 3 & 5 & -4 \end{bmatrix}$$

- A matrix is called triangular if it is either upper triangular or lower triangular.

Theorem

The determinant of a triangular matrix is equal to the product of its diagonal entries.

Example

Compute the determinant of $A = \begin{bmatrix} -2 & 2 & 0 & 3 & 1 \\ 0 & 2 & -5 & 7 & -5 \\ 0 & 0 & 3 & 4 & -4 \\ 0 & 0 & 0 & 9 & -4 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}$

One can use row or column expansions, but it is easier to note that the matrix A is upper triangular, so

$$\det A = (-2)(2)(3)(9)(-1) = 108$$

Theorem

The determinant of a triangular matrix is equal to the product of its diagonal entries.

Reason :

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{vmatrix}$$

$$= a_{11} \begin{vmatrix} a_{22} & a_{23} & a_{24} \\ 0 & a_{33} & a_{34} \\ 0 & 0 & a_{44} \end{vmatrix} - 0 \begin{vmatrix} a_{12} & a_{13} & a_{14} \\ 0 & a_{33} & a_{34} \\ 0 & 0 & a_{44} \end{vmatrix} + 0 \begin{vmatrix} a_{12} & a_{13} & a_{14} \\ a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{44} \end{vmatrix} - 0 \begin{vmatrix} a_{12} & a_{13} & a_{14} \\ a_{22} & a_{23} & a_{24} \\ 0 & a_{33} & a_{34} \end{vmatrix}$$

$$= a_{11}\left(a_{22} \begin{vmatrix} a_{33} & a_{34} \\ 0 & a_{44} \end{vmatrix} - 0 \begin{vmatrix} a_{23} & a_{24} \\ 0 & a_{44} \end{vmatrix} + 0 \begin{vmatrix} a_{23} & a_{24} \\ a_{33} & a_{34} \end{vmatrix}\right) = a_{11}a_{22}a_{33}a_{44}.$$

Summary: determinants and cofactor expansion

(i,j)-Cofactors

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

- Suppose A_{ij} denotes the (i,j)-minor of A. The (i,j)-cofactor of A is:

$$C_{ij} = (-1)^{i+j} \det A_{ij}$$

The sign $(-1)^{i+j}$ is determined by the following checkerboard pattern of signs :

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \end{bmatrix}$$

E.g., $C_{11} = \det A_{11}$, $C_{12} = -\det A_{12}$, $C_{32} = -\det A_{32}$, etc.

Determinant of an $n \times n$ matrix

Suppose $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$ is an $n \times n$ matrix.

- **Cofactor expansion across the p-th row :**

$$\det A = a_{p1}C_{p1} + a_{p2}C_{p2} + \cdots + a_{pn}C_{pn}$$

- **Cofactor expansion across the q-th column :**

$$\det A = a_{1q}C_{1q} + a_{2q}C_{2q} + \cdots + a_{nq}C_{nq}$$